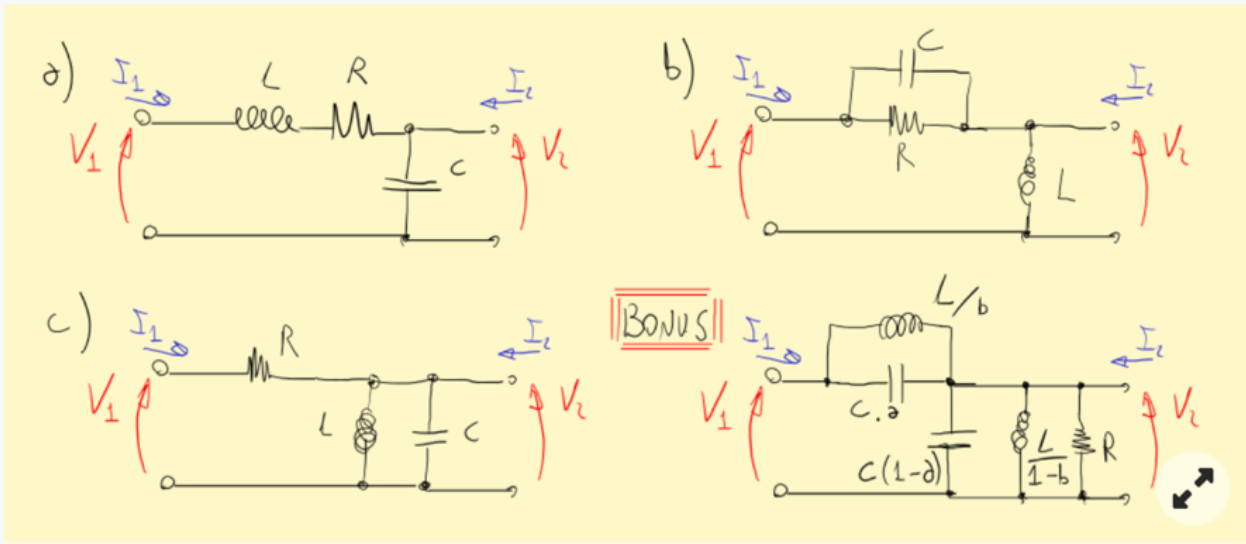
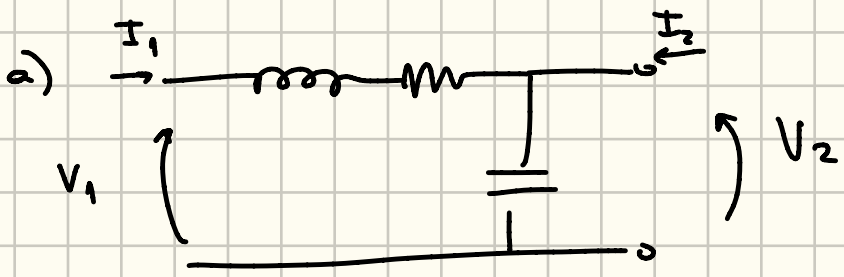


1. Hallar analíticamente la función de transferencia $H(s) = \frac{V_o(s)}{V_i(s)}$ para $\omega_0 = 1$, R = 1 Ω y $Q = \frac{\sqrt{2}}{2}$.
2. Calcular y graficar la respuesta de módulo y fase, indicando la respuesta asintótica donde corresponda.
3. Responda: ¿De qué tipo de filtro se trata? ¿Cuál es el ancho de banda del filtro?



$H(s) = \frac{V_o(s)}{V_i(s)}$ $\quad \omega_0 = 1$, $R = 1\,\Omega$ y $Q = \frac{\sqrt{2}}{2}$

$H(s) = \frac{Z_C}{Z_R + Z_C + Z_L}$ $\quad V_o = V_{in} \cdot \frac{Z_{solido}}{Z_{total}}$

$H(s) = \frac{1/sC}{R + \frac{1}{sC} + sL}$

$= \frac{1}{sCR + 1 + s^2CL}$ $\quad \omega_0^2 = \frac{1}{LC}$

$L = \frac{1}{C}$ $\quad \omega_0 \cdot \frac{L}{R} = Q$

$L = \frac{\sqrt{2}}{2}$

$C = \sqrt{2}$

$H(s) = \frac{1}{CL\left(s^2 + \frac{sR}{L} + \frac{1}{CL}\right)}$

$s = j\omega$

$s^2 = -\omega^2$

$H(j\omega) = \frac{1}{CL\left(-\omega^2 + j\omega\frac{R}{L} + \frac{1}{CL}\right)}$

$|H(j\omega)| = \frac{1/CL}{\sqrt{\left(\frac{1}{CL} - \omega^2\right)^2 + \left(\frac{\omega R}{L}\right)^2}}$

$\angle H = 0 - tg^{-1}\left(\frac{\frac{\omega R}{L}}{\frac{1}{CL} - \omega^2}\right)$

$|H(j\omega)| = \frac{1}{\sqrt{(1-\omega^2)^2 + 2\omega^2}}$

$\frac{1}{CL} = \frac{1}{\frac{\sqrt{2}}{2} \cdot \sqrt{2}} = 1$

$\frac{R}{L} = \frac{1}{\frac{\sqrt{2}}{2}} \cdot \sqrt{2}$

$= \sqrt{2}$

$$= \frac{1}{\sqrt{1 - 2\omega^2 + \omega^4 + 2\omega^2}}$$

$$|H(j\omega)| = \frac{1}{\sqrt{\omega^4 + 1}}$$

$$|H(j\omega)|_{dB} = 20 \log_{10}\left(\frac{1}{\sqrt{1+\omega^4}}\right)$$

$$= -10 \log_{10}(1+\omega^4)$$

si $\omega^4 = 0 \rightarrow |H|_{dB} = 0$

$\omega^4 >> 1 \rightarrow |H|_{dB} = -10 \log_{10}(\omega^4)$ ($\omega \rightarrow \infty$)

$$= -40 \log_{10}(\omega) \rightarrow \text{recta pend de } -40 dB/dec$$

$\omega = 1 \quad |H|_{dB} = -10 \log_{10}(2)$

$$= -3 \text{ dB}$$

$$\measuredangle H = 0 - tg^{-1}\left(\frac{\sqrt{2}\,\omega}{1-\omega^2}\right)$$

$\omega \rightarrow 0 \quad \measuredangle H = 0 - tg^{-1}(0)$

$= 0$

$\omega = 1 \quad \measuredangle H = -tg^{-1}\left(\frac{\sqrt{2}}{0}\right)$ → se rompe, para $\times \frac{\pi}{2}$

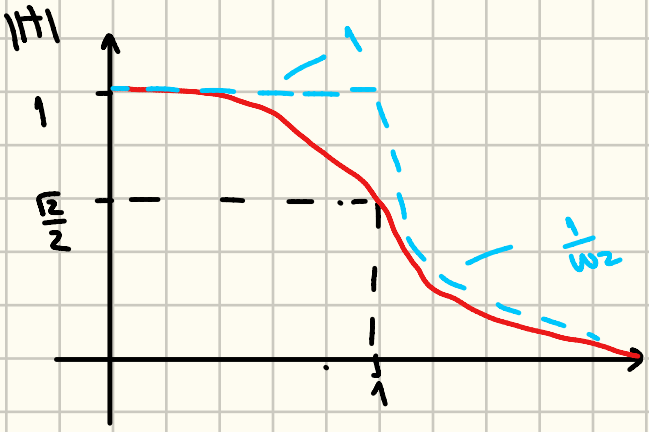
$\underbrace{\quad}_{-\pi/2}$ $(0, \sqrt{2})$ (abre eje imaginario $\oplus$, $1 - \frac{\pi}{2}$)

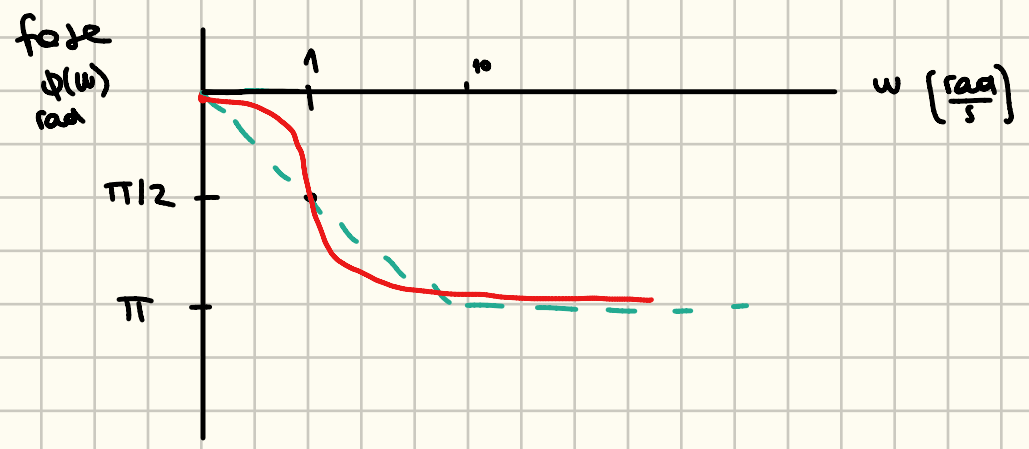
$\omega \rightarrow \infty \quad \measuredangle H = -tg^{-1}\left(\frac{\sqrt{2}\,\omega}{1-\omega^2}\right)$

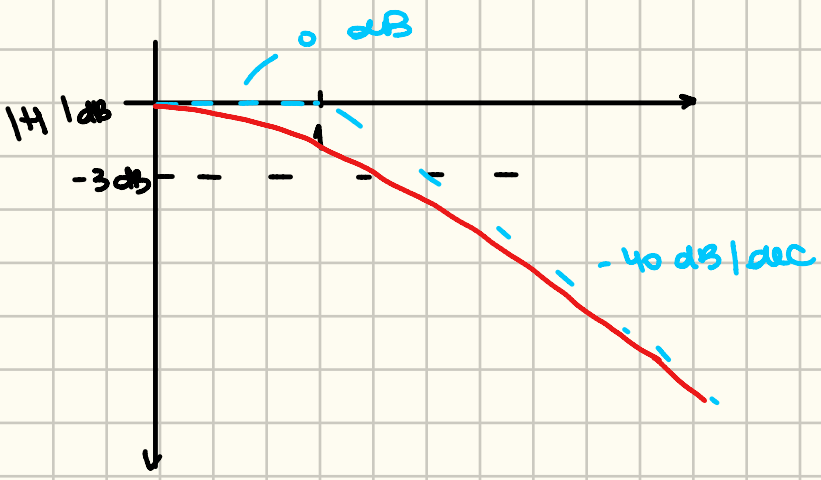
$\downarrow$ ω^2 gana

2° cuadrante, pegado eje IR $\ominus$

$\underbrace{\quad}_{-\pi}$







$\sqrt{\omega_0} = \sqrt{1}$

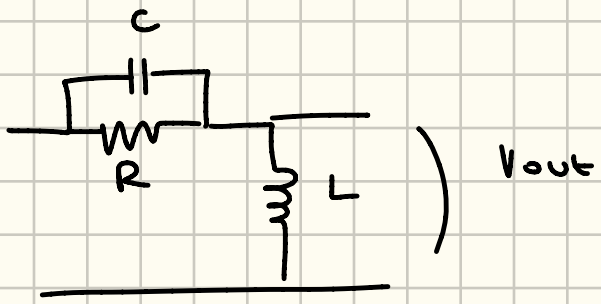
frec. de corte : $\omega = 1 \frac{rad}{s}$

$\underbrace{\quad}_{-3dB \text{ ó } H = \frac{1}{\sqrt{2}}}$

filtro pasabajo de orden 2

$BW = \omega_c = 1 \frac{rad}{s}$ deja pasar desde continua hasta la frec. de corte ($\omega_c = 1$ rad/s)

b)



$$H(s) = \frac{V_o}{V_{in}}$$

$$V_o = V_{in} \cdot \frac{z_{salida}}{z_{total}}$$

$$z_1 = z_C \parallel z_R = \frac{z_C \cdot z_R}{z_C + z_R} = \frac{R \cdot 1/sC}{\left(R + \frac{1}{sC}\right)} = \frac{R}{sCR+1}$$

$$H(s) = \frac{sL}{sL + \frac{R}{sCR+1}} = \frac{sL(sCR+1)}{sL(sCR+1) + R} = \frac{s^2LCR + sL}{s^2LCR + sL + R} = \frac{s^2 + sL/LCR}{s^2 + sL/LCR + R/LCR}$$

$$H(s) = \frac{s^2 + s/CR}{s^2 + s/CR + 1/LC}$$

$$w_o^2 = \frac{1}{LC} \qquad \frac{w_o}{Q} = \frac{1}{RC}$$

$$L = \frac{1}{C} \qquad C = Q$$

$$C = \frac{\sqrt{2}}{2}$$

$$L = \sqrt{2}$$

$$\frac{1}{CR} = \frac{1}{\frac{1}{\sqrt{2}}}$$

$$\frac{1}{LC} = \frac{1}{\frac{1}{\sqrt{2}} \cdot \sqrt{2}} = 1$$

$$H(s) = \frac{s^2 + \sqrt{2}s}{s^2 + \sqrt{2}s + 1}$$

$$H(jw) = \frac{-w^2 + jw\sqrt{2}}{-w^2 + jw\sqrt{2} + 1}$$

$$|H(jw)| = \frac{\sqrt{(-w^2)^2 + w^2 \cdot 2}}{\sqrt{(1-w^2)^2 + w^2 \cdot 2}} = \frac{\sqrt{w^4 + w^2 \cdot 2}}{\sqrt{(1-w^2)^2 + w^2 \cdot 2}}$$

$$|H(jw)| = \frac{w\sqrt{w^2+2}}{\sqrt{(1-w^2)^2 + w^2 \cdot 2}}$$

$$|H(jw)| = \frac{w\sqrt{w^2+2}}{\sqrt{w^4+1}}$$

$$|H|_{dB} = 20 \log_{10} |H| =$$

$w \to 0 \quad |H| = \sqrt{2} \cdot w \to 0$

$w = 1 \quad |H| = \frac{\sqrt{3}}{\sqrt{2}} = 1{,}22$

$w \to \infty \quad |H| = 1$

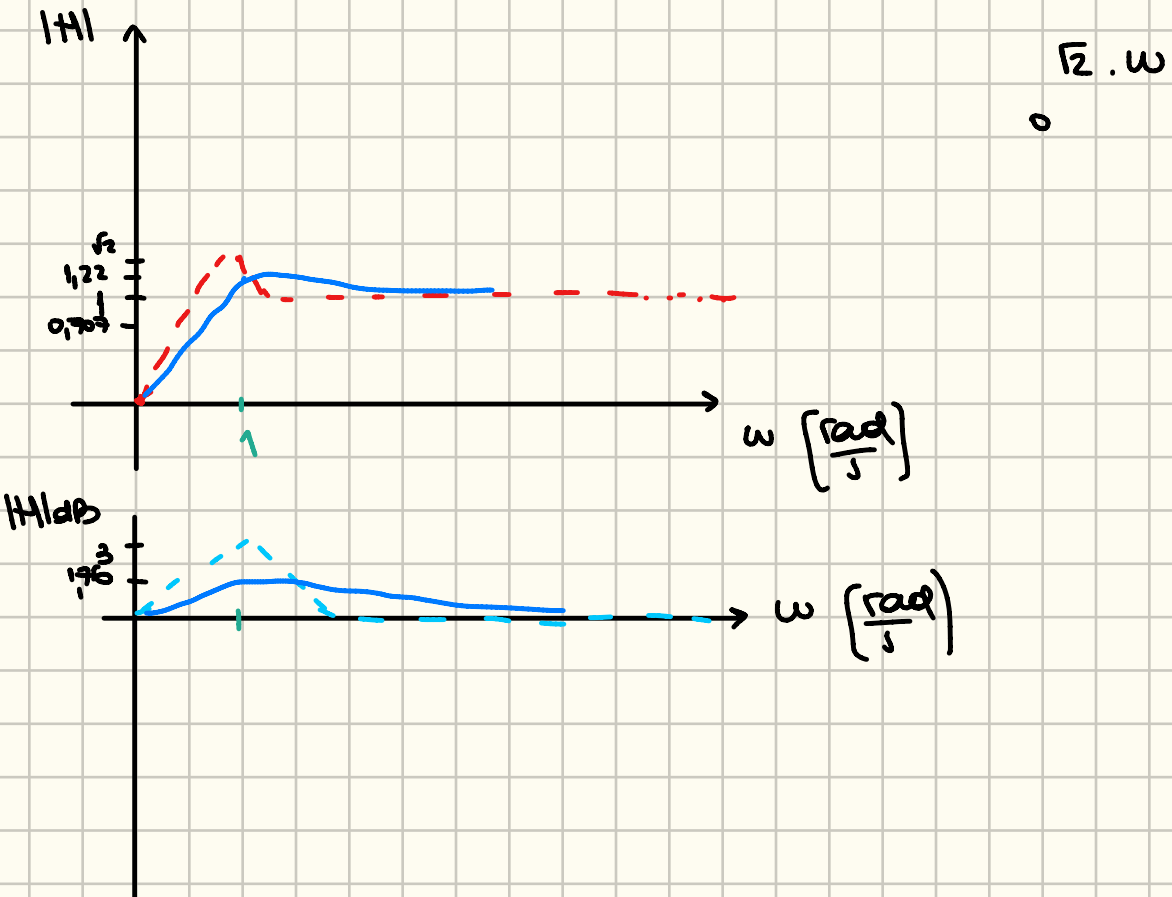
$\left(\frac{\sqrt{w^4}}{\sqrt{w^4}}\right)$ opok

$$|H|_{dB} = 20 \log_{10} |\omega\sqrt{\omega^2+2}| - 20 \log_{10} \sqrt{\omega^4+1}$$

$\omega \to 0$ $\quad |H|_{dB} = 20 \log_{10} (\sqrt{2}.\omega) \approx 3{,}01 \xrightarrow{\omega \to 0} 0$

$\omega = 1$ $\quad |H|_{dB} = 20 \log_{10} (1{,}22) = 1{,}76\ dB$

$\omega \to \infty$ $\quad |H|_{dB} = 20 \log_{10}(1) = 0$



$\sqrt{2}.\omega$

0

$$\measuredangle H = tg^{-1}\left(\frac{\omega\sqrt{2}}{-\omega^2}\right) - tg^{-1}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right)$$

corrijo

$\omega \to 0$

$$\measuredangle H = tg^{-1}\left(\frac{\sqrt{2}}{-\omega}\right) - tg^{-1}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right)$$

$(\pi - \frac{\pi}{2})$ $\quad \to \infty$ $\quad$ (+) $\quad$ 0

$$\measuredangle H = \frac{\pi}{2} - 0$$

→ muy chico : valor Re (+)

$$\measuredangle H = \frac{\pi}{2} \quad \to \quad \omega \to 0$$

$\omega = 1$ $\quad \measuredangle H = tg^{-1}(-\sqrt{2}) - tg^{-1}\left(\frac{\sqrt{2}}{0}\right) = 0{,}611\ rad$

$0{,}955\ rad$ $\quad \frac{\pi}{2}$

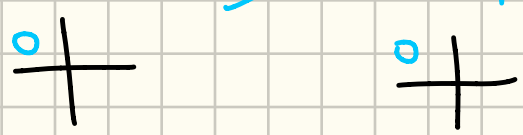
para Real en ⊖ → $\pi - 0{,}955 = 2{,}186\ rad$

0

↓

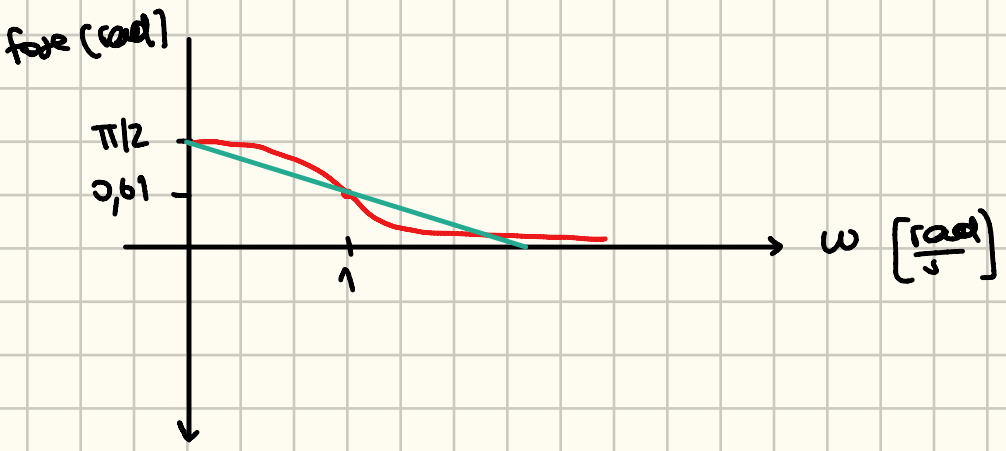
corregir

$\omega \to \infty$



$$\measuredangle H = \tan^{-1}\left(\frac{\sqrt{2}}{-\omega}\right) - \tan^{-1}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right)$$

$$\measuredangle H = \pi - 0 - \pi - 0 = 0$$



filtro Pasa-Altos de 2° orden

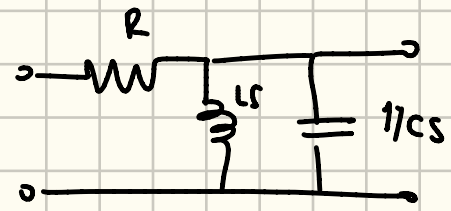
$\omega = 0 \rightarrow$ ganancia es cero: $|H(0)| = 0$

$\omega \to \infty \rightarrow$ ganancia tiende a 1 (0 dB)

- Deja pasar todas las frec. superiores a $\omega_c = 1$ rad/s

$$BW = [1\ \text{rad/s}, \infty)$$

c)



$$H(s) = \frac{V_{out}}{V_{in}}$$

$$H(s) = \frac{sL}{sL + s^2CLR + R}$$

$$H(s) = \frac{s/CR}{s^2 + s.\frac{1}{CR} + \frac{1}{LC}}$$

$\omega_0 = 1 \qquad R = 1\ \Omega \qquad \omega = \frac{\sqrt{2}}{2}$

$$\omega_0^2 = \frac{1}{LC} \qquad \frac{1}{RC} = \frac{\omega_0}{Q}$$

$$L = \frac{1}{C} \qquad \frac{Q}{\omega_0 . R} = C$$

$$\frac{\sqrt{2}}{2} = C \qquad L = \sqrt{2} \qquad \frac{1}{RC} = \frac{1}{\frac{\sqrt{2}}{2}} = \sqrt{2}$$

z salida — $z_s \,//\, z_c$

$$z_2 = \frac{Ls \,.\, 1/Cs}{Ls + \frac{1}{Cs}} = \frac{sL}{s^2CL + 1}$$

$$V_{out} = V_{in} . \frac{\frac{sL}{s^2CL+1}}{\frac{sL}{s^2CL+1} + R}$$

$$= V_{in} . \frac{sL}{sL + s^2CLR + R}$$

$$H(s) = \frac{\sqrt{2}s}{s^2 + \sqrt{2}s + 1} \Bigg|_{s=j\omega}$$

$$H(j\omega) = \frac{\sqrt{2}j\omega}{-\omega^2 + j\omega\sqrt{2} + 1}$$

$$|H(j\omega)| = \frac{\sqrt{2}\omega}{\sqrt{(1-\omega^2)^2 + \omega^2 \cdot 2}} = \frac{\sqrt{2}\omega}{\sqrt{\omega^4 + 1}}$$

$$|H(j\omega)|_{dB} = 20 \log_{10}\left(\frac{\sqrt{2}\omega}{\sqrt{\omega^4+1}}\right)$$

$\omega \to 0$ $\quad H(j\omega) \to \sqrt{2}.\omega$

$\omega \to 0$ $\quad |H(j\omega)|_{dB} \to 20 \log(\sqrt{2}.\omega) \to -\infty dB$ ($\omega \to 0$)

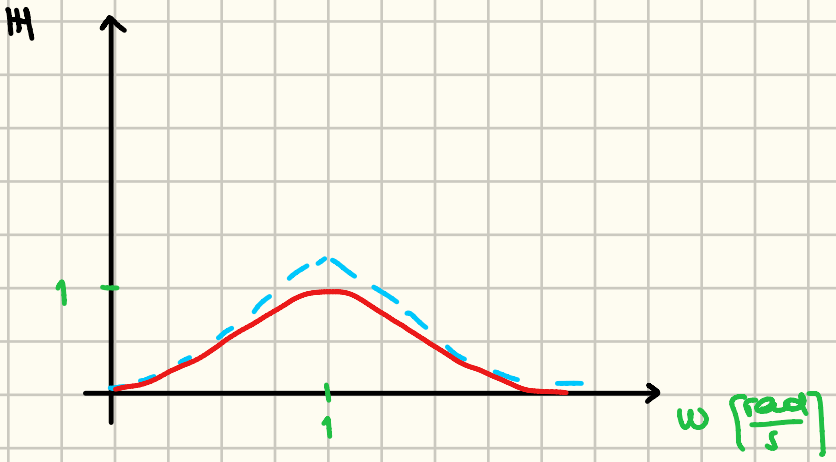
$\omega = 1$ $\quad H(j\omega) = \frac{\sqrt{2}}{\sqrt{2}} = 1$

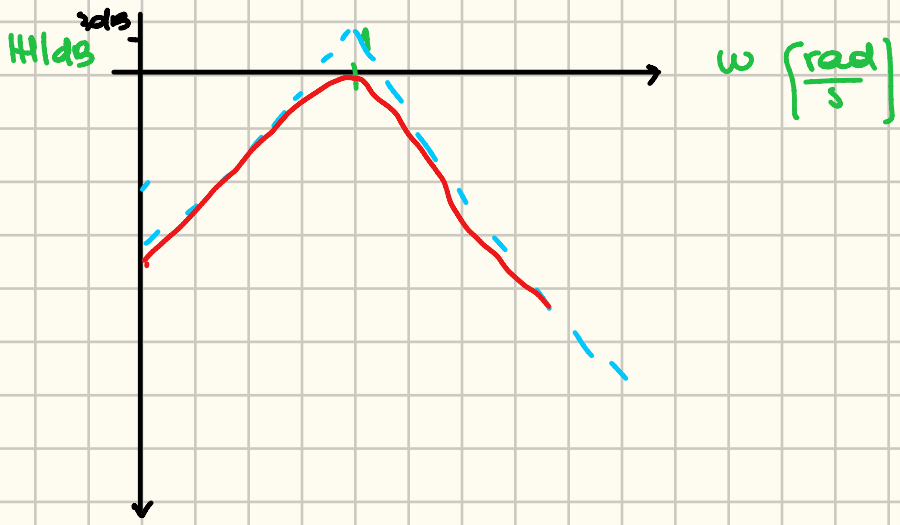
$\omega = 1$ $\quad |H(j\omega)|_{dB} = 20 \log(1) \to 0 dB$

$\omega \to \infty$ $\quad H(j\omega) \to \frac{\sqrt{2}\omega}{\omega^2} = \frac{\sqrt{2}}{\omega} \to$ Recta

$\sqrt{\omega^4+1} \approx \omega^2$

$\omega \to \infty$ $\quad |H(j\omega)|_{dB} = 20 \log\left(\frac{\sqrt{2}}{\omega}\right) \to -\infty dB$ ($\frac{\sqrt{2}}{\omega} \to 0$)





$$\measuredangle H = tg^{-1}\left(\frac{\sqrt{2}\omega}{0}\right) - tg^{-1}\left(\frac{\omega\sqrt{2}}{(1-\omega^2)^2}\right)$$

100% imaginario

$\frac{\pi}{2}$

$$\measuredangle H = 90° - tg^{-1}\left(\frac{\omega\sqrt{2}}{1-\omega^2}\right)$$

$\omega \to 0 \qquad \varphi H = \frac{\pi}{2} - tg^{-1}\left(\frac{\omega f_2}{1-\omega^2}\right) = \frac{\pi}{2} - 0 = \frac{\pi}{2}$

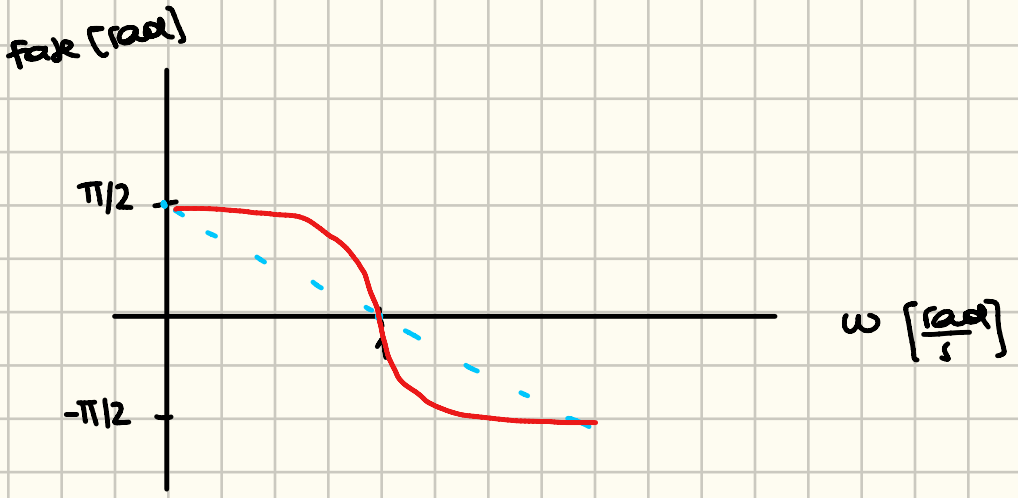
$\hookrightarrow \hookrightarrow 0$

$\omega = 1 \qquad \varphi H = \frac{\pi}{2} - tg^{-1}\left(\frac{f_2}{0}\right) = \frac{\pi}{2} - \frac{\pi}{2} = 0$

$\hookrightarrow \infty$

$\omega \to \infty \qquad \varphi H = \frac{\pi}{2} - tg^{-1}\left(\frac{\omega f_2}{1-\omega^2}\right) = \frac{\pi}{2} - \pi = -\frac{\pi}{2}$

para $1-\omega^2$ ↓ en 2° cuadrante



filtro pasa banda de 2° orden

$BW = \frac{\omega_0}{Q} = \frac{1}{\frac{\sqrt{2}}{2}} = \sqrt{2} = 1{,}41 \frac{rad}{seg}$